

Genetic food intolerances (626 foods)

Genetic predisposition predicted for 626 frequent consumption foods worldwide, by category
Report generated at 2 of October 2026 (local time)

Genetic basis: Raw DNA of a genotyping chip (GRCh38, 684 967 SNPs) + local imputation (Beagle 5.5, full genome reference panel 30x; only genotypes with DR2 ≥ 0.8). Genetic data did not leave the local computer. Only genetics: no analyses or exams were used.

Mechanisms evaluated: 28 mechanisms (lactase, celiac disease, hereditary intolerance to fructose, saccharase-isomaltase, histamine, caffeine, salt, alcohol, liver × sugar, glycemic response, iron, smoking detoxification, favism, oxalate, purines, PKU, saturated fat, bitter taste and coriander, among others), with ~140 variants and ClinVar sweep of ~130 food metabolism genes.

List of foods: Foods most consumed worldwide (Food Sheets products from FAO unfolded in concrete foods, food composition tables and frequent processed products), without duplicates.

Important: is a genetic reading of predisposition, not a diagnosis. Genetics explains only one part of the intolerances; IgE allergies and FODMAP sensitivity are not predictable by genes. All indications are suggestion (not applied).

red avoid / high probability **orange** limit, dose-dependent effect **yellow** monitoring / possible sensitivity. On request, the report shows only predisposed foods; the rest are counted but not listed.

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1. Executive summary

Class	Food No.	%
Avoiding high probability of intolerance	0	0.0%
Limit (dose dependent)	140	22.4%
Monitoring / possible sensitivity	162	25.9%
No genetic predisposition (gene evaluated without alteration, non-determinable or no known genetic binding)	324	51.8%
Total analyzed	626	100%

324 of the 626 verified foods do not show genetic predisposition (are not listed in this report).

Fast reading. No food is red: there is no genetic intolerance "classical". Orange and yellow colors come from dose-dependent effects, not true intolerance:

- **Free sugar and excess fructose** (Orange; sugar, honey, syrups, soft drinks, juices, sweets and pastries): PNPLA3 I148M + MBOAT7 + Pi*MZ probably make the liver more sensitive to high doses of sugar; the protective HSD17B13 attenuates. The whole fruit does not enter here.
- **Sal** (Orange in very salty: sausages, ham, very salty cheeses, salty candies, soy sauce, cube juices, snacks): AGT, ADD1, CYP11B2, GNB3 and ACE in salt sensitivity profile.
- **Coffee** (Orange in coffee and energy drinks; yellow in tea, glue and black chocolate): CYP1A2 *1F A/C (normal to intermediate elimination) + ADORA2A C/C (less sensitive); the limit comes from sleep (insomnia p89).
- **Alcohol** (Orange in all alcoholic beverages): no flushing (ALDH2 and ADH1B normal), but the genetic liver (PNPLA3 + Pi*MZ) increases susceptibility.
- **Histamine** (yellow: cured cheeses, sausages, fish in canned or smoked, fermented, wine, vinegar, spinach, eggplant, avocado): DAO with 2 heterozygous variants + HNMT Thr105Ile heterozygous. Weak evidence.
- **Sucrose** (yellow; reinforces sweet orange): SI Val15Phe heterozygous, with slightly lower saccharase activity.
- **Refined hydrates** (yellow: white bread, white rice, potato, refined flour): TCF7L2, FTO, MTNR1B and FABP2 heterozygotes — intermediate glycemic response, worst at night.
- **Other yellows:** viscera and morcella (ferro heme; HFE H63D heterozygous), carbonized and smoked meat (GSTM1 probable null + GSTA1*B/*B + NAT2 slow), fortified foods (iron and folic acid; HFE + compound MTHFR), liquor/regaliz (salt sensitivity) and coriander (OR6A2: only possible soap flavor).

Good genetic news: Lactose tolerant (LCT, 2 SNPs), without genetic risk of celiac disease (without DQ2/DQ8), without hereditary intolerance to fructose (ALDOB), without favism (G6PD), without PKU (PAH), typical response to saturated fat (APOA2, APOE ε3/ε3, LIPC), low predisposition to high uric acid (normal ABCG2), without genetic hyperoxaluria (AGXT), intermediate taster of bitter in the crucifers (which with null GSTM1 give even more benefit) and secretor FUT2.

What this work adds to what was already known

- **Saccharase-isomaltase (SI) Val15Phe heterozygous** (rs9290264 C/A, chip): a common variant of reduced function associated with irritable bowel syndrome and symptoms with sucrose. No G1073D. *New.*
- **ALDOB:** A149P, A174D and N334K absent → hereditary intolerance to fructose very unlikely. *New.*
- **PAH:** 9 frequent variants of PKU absent (before only R408W). *New.*
- **Lactase confirmed by 2.o SNP** (−22018, rs182549) and without other persistence alleles. DQ2.2 also absent. *Reinforcement.*
- **Sal:** The high sensitivity profile of the previous interpretation is explained by ADD1 Gly460Trp, CYP11B2 −344 T/T, GNB3 825T and AGT M235T. *Fine new.*

- **Cafein:** rs2472297 and rs2470893 C/C (CYP1A1/CYP1A2, without the fast elimination allele) are added to CYP1A2 *1F A/C and ADORA2A C/C. *Fine new.*
- **Coriander:** OR6A2 rs72921001 C/A, 1 allele associated with soap flavor (small effect). *New.*
- **AGXT** Heterozygous haplotype (P11L + I340M) without G170R: no isolated clinical relevance. **APOA2** –265 not C/C and **LIPC** –514 C/C: typical response to saturated fat. *New.*
- **ClinVar in ~130 genes of food metabolism** (chip + imputation): no new pathogenic variant; only Pi*Z already known. *New.*
- Confirmed: DAO/HNMT slightly reduced, G6PD normal, ALDH2/ADH1B normal, FUT2 secretory, GALT Duarte heterozygote (without relevance), HFE H63D heterozygote, MTHFR compound, GSTM1 null probable.

Non-genetic notes (do not enter the colors): Grapefruit and pomelium interact with drugs metabolized by CYP3A4; unfiltered coffee (embolus, boiled) rises LDL by the coffee; vitamin K of green leaves only matters if one day there is anticoagulation with coumarins (VKORC1 T/T + CYP2C9 *1/*3).

2. Mechanisms and genes evaluated

Mechanism	Genes	Genotype (strand +, GRCh38)	Interpretation	Evidence	Trust
Free sugar/fructose x liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	PNPLA3 I148M C/G; MBOAT7 C/T; Probable Pi*MZ; HSD17B13 protector (attenuous)	High doses of free sugar/fructose increase liver fat in patients with 148M	Moderate	chip + imputation
Coffee	CYP1A2, ADORA2A	CYP1A2 *1F A/C (normal to intermediate elimination; rs2472297 and rs2470893 C/C, without the fast allele) + ADORA2A rs5751876 C/C	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p89): dose and time dependent effect	Moderate	chip
Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	AGT M235T A/G; ADD1 G460W G/T; CYP11B2 -344 T/T; GNB3 825 C/T; ACE (rs4343 G/A)	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Moderate-weak	chip
Alcohol	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7	ALDH2 G/G and ADH1B C/C (without flushing); PNPLA3 I148M; Probable Pi*MZ; MBOAT7 C/T	Normal acute tolerance, but higher hepatic susceptibility to alcohol	Moderate-strong	chip + imputation
Sucrose (sucrose-isomaltase)	SI	Val15Phe (rs9290264 C/A) heterozygous; without G1073D	Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate-weak	chip
Food Histamine	AOC1 (DAO), HNMT	DAO Thr16Met C/T + His645Asp C/G; HNMT Thr105Ile C/T (all heterozygous); without -691T or Ser332Phe	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Weak	chip
Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	TCF7L2 rs7903146 C/T + rs12255372 G/T; FTO het; MTNR1B C/G; FABP2 Ala54Thr het	Intermediate glycemic response; worst at night (MTNR1B)	Moderate-weak	chip + imputation
Iron heme	HFE, TMPRSS6	HFE H63D C/G (without C282Y or S65C); TMPRSS6 A/G	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Weak	chip
Fortified foods (iron, folic acid)	HFE, MTHFR	HFE H63D C/G; MTHFR C677T G/A + A1298C T/G	Iron added and synthetic folic acid: prefer natural sources	Weak	chip
Smoked/carbonized meat (HAP, heterocyclic amines)	GSTM1, GSTA1, NAT2	Probable null GSTM1; GSTA1 *B/*B; NAT2 *5/*5 (slow)	Weakest II phase detoxification for HAP and heterocyclic amines	Weakness	chip + inference (GSTM1 is CNV)
Corianders (soap flavor)	OR6A2	rs72921001 C/A (1 allele "sabão")	A slightly higher probability of finding the corianders with flavor (only flavor)	Weak	chip
Liquorice/regaliz (glycirrhizal)	CYP11B2, AGT, ADD1	CYP11B2 -344 T/T + salt sensitivity profile	Glycirrhizin retains sodium; more relevant with salt sensitivity	Weak	chip
Lactose	LCT/MCM6	-13910 C/T (rs4988235 G/A) + -22018 G/A (rs182549 C/T): 1 persistence allele	Lactase persistence (dominant): lactose tolerant	Strong	chip (2 SNPs in agreement)
Gluten — Celiac Disease	HLA-DQ2.5/DQ2.2/DQ8	rs2187668 C/C; rs7454108 T/T; negative DQ2.2 tags; HLA imputed without DQ2/DQ8; PGS celiac p3,9	Without the haplotypes required for celiac disease (negative predictive value ≈ 99%)	Strong	chip + imputation
Hereditary intolerance to fructose	ALDOB	A149P, A174D and N334K absent	Without the variants that cause most cases	Strong	chip
Favism	G6PD	Without A-, without c.563C>T (normal hemizigote)	No genetic risk of hemolysis with favae	Strong	chip
Saturised fat	APOA2, APOE, LIPC	APOA2 -265 T/C (not C/C); APOE ε3/ε3; LIPC -514 C/C	Typical response to saturated fat	Moderate-weak	chip + imputation

Mechanism	Genes	Genotype (strand +, GRCh38)	Interpretation	Evidence	Trust
Oxalate	AGXT	P11L C/T + I340M A/G (common haplotype, heterozygous); G170R absent	No relevant genetic predisposition to hyperoxaluria	Moderate	chip
Purines (uric acid)	ABCG2, SLC2A9, SLC22A12	ABCG2 Q141K and Q126X absent; SLC2A9 heterozygous; SLC22A12 T/C	Low predisposition to hyperuricemia (without the most effective ABCG2 allele)	Moderate	chip + imputation
Bitter / crucifers	TAS2R38, GSTM1	TAS2R38 heterozygous in the 3 SNPs (PAV/AVI probable); Probable null GSTM1	Intermediate tester; with zero GSTM1 isothiocyanates are longer (benefit)	Moderate	chip
Phenylalanine (aspartame)	PAH	R408W, IVS12+1, IVS10-11, R261Q, R158Q, P281L, I65T, A403V, L48S absent	Without PKU (aspartame is restricted only in PKU)	Strong	chip
Galactose	GALT	N314D (Duarte) A/G heterozygous	Portador Duarte: no clinical relevance	Moderate	chip
Salivary starch	AMY1	number of non-measurable copies	Non-determinable (CNV); SI Val15Phe can slightly reduce isomaltase	—	not determinable
FODMAP / polyols/fructose in excess	—	without validated genetic test	Not determinable by genetics	—	not determinable
Allergy IgE	HLA-DQB1, IL4R, FLG	IL4R Q576R A/G; FLG possible unconfirmed LoF; HLA rs927596 C/T	IgE food allergy not predictable by genetics (atopic context only)	Weak	partial
Mastocytes / tryptase	TPSAB1	number of non-measurable copies	Hereditary alpha1-tryptasemia not determinable	—	not determinable
Alpha-gal (red meat)	ABO	The probable group	Alpha-gal syndrome depends on tick bites; genetics hardly predicts	Very weak	chip + imputation
Vitamin B12 absorption/microbiome	FUT2	W154X G/A (secretor)	Secretor: typical absorption	Moderate	chip
Omega-3 / fatty acids	FADS1/2	rs174546 C/C, rs174537 G/G	Efficient conversion of ALA/LA	Moderate	chip

Colors of this table: orange/yellow : mechanism with predisposition; green = gene evaluated without predisposition; gray = non-determinable. The "informative" mechanisms (releasers of histamine, blocking DAO by tea, coffee, grapefruit, vitamin K, iron in normal portions) do not give color to any food.

3. Foods with predisposition, by category

In each category only red, orange or yellow foods appear, ordered by gravity. The last line of each category indicates how many foods were verified without genetic predisposition.

3.1 Dairy products — 42 verified

limit: 7 · surveillance: 14 · without genetic predisposition: 21

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Natal ice cream Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Condensed milk Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Blue cheese (roquefort, gorgonzola) Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Feta cheese Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Sliced cheese Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak <i>chip</i>
Halloumi cheese Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak <i>chip</i>
Parmesan cheese Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Sweet rice Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Refined hydrates / high glycemic index	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; TCF7L2, FTO, MTNR1B, FABP2	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose · Intermediate glycemic response; worst at night (MTNR1B)	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate <i>chip + imputation</i>
Sugary yogurt with fruit Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Sugary liquid yogurt Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Milk Kefir Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Chocolate milk Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Milk cream / milk pudding Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Potatoes to beat with sugar Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Brie cheese/camembert Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Cheddar cheese Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Cow curated cheese Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Cure goat cheese Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Cheese of buttery sheep Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Young semi-hard cheese Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Emmental cheese Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>

3.2 Starch and derivatives — 55 verified foods

limit: 2 · surveillance: 24 · without genetic predisposition: 29

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Chocolate balls Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Sweetened breakfast cereals Limit	Free sugar/fructose × liver Refined hydrates / high glycemic index Sucrose (sucrose-isomaltase) Fortified foods (iron, folic acid)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; TCF7L2, FTO, MTNR1B, FABP2; SI; HFE, MTHFR	High doses of free sugar/fructose increase liver fat in patients with 148M · Intermediate glycemic response; worst at night (MTNR1B) · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose · Iron added and synthetic folic acid: prefer natural sources	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Prefer integral versions, moderate portions, with protein or fiber and early in the day. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer versions without iron or folic acid added.	Moderate <i>chip + imputation</i>
Needle rice Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
White rice Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Carolina rice Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Baguette Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak <i>chip + imputation</i>
Rice cake/gallet Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Maria? Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Refined hydrates / high glycemic index	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; TCF7L2, FTO, MTNR1B, FABP2	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose · Intermediate glycemic response; worst at night (MTNR1B)	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate <i>chip + imputation</i>
Water and salt bubbles Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak <i>chip + imputation</i>
Croissant Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Cuscouis Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Rice flour Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Corn flour/polenta Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Corn flakes Monitor	Refined hydrates / high glycemic index Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Fortified foods (iron, folic acid) Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; HFE, MTHFR; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose · Iron added and synthetic folic acid: prefer natural sources · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer versions without iron or folic acid added. Count in daily salt limit (< 5 g).	Moderate-weak <i>chip + imputation</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Granule Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Sugar muesli Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Papo-seco/carcass Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak <i>chip + imputation</i>
Pizza Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak <i>chip + imputation</i>
White wheat bread Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak <i>chip + imputation</i>
White bread Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak <i>chip + imputation</i>
Corn bread (broa) Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Mixture bread (wheat and rye) Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Pita/naan bread Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak <i>chip + imputation</i>
Gluten-free bread Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Wheat semole Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Toastes Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>

3.3 Tubercules and roots — 12 verified foods

limit: 1 · surveillance: 6 · without genetic predisposition: 5

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Potatoes of package Limit	Salt (genetic sensitivity to salt) Refined hydrates / high glycemic index	AGT, ADD1, CYP11B2, GNB3, ACE; TCF7L2, FTO, MTNR1B, FABP2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Intermediate glycemic response; worst at night (MTNR1B)	Small and infrequent portions; prefer versions with less salt. Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip</i>
Roast potato Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Cooked potato Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Fried potato Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Cassava flour / tapioca Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Potato starch Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Potato puree Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>

3.4 Legumes and derivatives — 25 verified foods

limit: 1 · surveillance: 3 · without genetic predisposition: 21

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Miso Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Natto Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Tempeh Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Lupins Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>

3.5 Vegetables, mushrooms and algae — 67 verified foods

limit: 3 · surveillance: 13 · without genetic predisposition: 51

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Olives in brine Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Kimchi Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Pickles / cucumber in vinegar Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Avocado Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Wakame / kombu Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Canned garlic Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Beringela Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Chucrutz Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Coriander (fresh) Monitor	Corianders (soap flavor)	OR6A2	A slightly higher probability of finding the corianders with flavor (only flavor)	It only affects the taste; without implication for health.	Weak <i>chip</i>
Canned mushrooms Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Tomato concentrate Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Spinach Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Frozen spinach Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Gaspacho Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Ketchup Monitor	Food Histamine Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose. Count in daily salt limit (< 5 g).	Weak <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Roasted peppers Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>

3.6 Fruits and derivatives — 58 verified foods

limit: 7 · surveillance: 12 · without genetic predisposition: 39

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Fruit jam/sweet Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Fruit in syrup Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Marmalade Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Fruit nectar Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Package orange juice Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate <i>chip + imputation</i>
Apple juice Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate <i>chip + imputation</i>
Grape juice Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate <i>chip + imputation</i>
Dry plums Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate <i>chip + imputation</i>
Sugar-dried cranberries Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Dehydrated banana Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate <i>chip + imputation</i>
Dry Apricots Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate <i>chip + imputation</i>
Dry figs Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate <i>chip + imputation</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Grapes Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate <i>chip + imputation</i>
Fruit puré (baby) Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate <i>chip + imputation</i>
Fruit smoothie Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate <i>chip + imputation</i>
Anasine juice Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate <i>chip + imputation</i>
Natural orange juice Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate <i>chip + imputation</i>
Grapefruit juice Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate <i>chip + imputation</i>
Dates Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate <i>chip + imputation</i>

3.7 Dry fruits and seeds — 25 verified foods

limit: 1 · surveillance: 3 · without genetic predisposition: 21

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Hazelnut and cocoa cream Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Peanuts roasted with salt Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Rice drink Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip + imputation</i>
Dry fruits with honey Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>

3.8 Meat and viscera — 32 verified foods

surveillance: 11 · without genetic predisposition: 21

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Well-done/carbonized grilled meat Monitor	Smoked/carbonized meat (HAP, heterocyclic amines)	GSTM1, GSTA1, NAT2	Weakest II phase detoxification for HAP and heterocyclic amines	Avoid carbonization; marinate and cook at moderate temperature.	Weakness <i>chip + inference</i> (GSTM1 is CNV)

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Meat in the grill (coal) Monitor	Smoked/carbonized meat (HAP, heterocyclic amines)	GSTM1, GSTA1, NAT2	Weakest II phase detoxification for HAP and heterocyclic amines	Avoid carbonization; marinate and cook at moderate temperature.	Weakness <i>chip + inference (GSTM1 is CNV)</i>
Heart Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak <i>chip</i>
Folding / tripe Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak <i>chip</i>
Roast chicken in the grill Monitor	Smoked/carbonized meat (HAP, heterocyclic amines)	GSTM1, GSTA1, NAT2	Weakest II phase detoxification for HAP and heterocyclic amines	Avoid carbonization; marinate and cook at moderate temperature.	Weakness <i>chip + inference (GSTM1 is CNV)</i>
Chicken liver Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak <i>chip</i>
Pork liver Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak <i>chip</i>
Cow liver Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak <i>chip</i>
Cow tongue Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak <i>chip</i>
Molasses Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak <i>chip</i>
Rim Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak <i>chip</i>

3.9 Charcutery and cured or smoked meats — 20 verified foods

limit: 17 · surveillance: 3 · without genetic predisposition: 0

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Alheira Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Smoked Bacon Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Bresaola Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Hot dog Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Meat chorizon Limit	Salt (genetic sensitivity to salt) Food Histamine Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Blood chorizon / morcela Limit	Salt (genetic sensitivity to salt) Food Histamine Iron heme	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; HFE, TMPRSS6	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Moderate portions; no iron supplementation or vitamin C in the same meal.	Moderate-weak <i>chip</i>
Farinheira Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Tongue Limit	Salt (genetic sensitivity to salt) Food Histamine Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Mortadela Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak <i>chip</i>
Paio / curated loin Limit	Salt (genetic sensitivity to salt) Food Histamine Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Pancetta Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Pastrami / smoked meat Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Curated ham Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Salami Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Salpicon Limit	Salt (genetic sensitivity to salt) Food Histamine Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Smoked sausages Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Smoked meat Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Braid Monitor	Salt (genetic sensitivity to salt) Fortified foods (iron, folic acid)	AGT, ADD1, CYP11B2, GNB3, ACE; HFE, MTHFR	Moderate-high salt: counting daily · Iron added and synthetic folic acid: prefer natural sources	Count in daily salt limit (< 5 g). Prefer versions without iron or folic acid added.	Moderate-weak <i>chip</i>
Turkey ham Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Liver paw Monitor	Iron heme Salt (genetic sensitivity to salt) Food Histamine	HFE, TMPRSS6; AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk) · Moderate-high salt: counting daily · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Moderate portions; no iron supplementation or vitamin C in the same meal. Count in daily salt limit (< 5 g). Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>

3.10 Fish — 42 verified foods

limit: 4 · surveillance: 12 · without genetic predisposition: 26

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Canned anchovy/salted anchovy Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Smoked herb Limit	Salt (genetic sensitivity to salt) Food Histamine Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Salted dry fish (others) Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Smoked salmon Limit	Salt (genetic sensitivity to salt) Food Histamine Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Fresh herb Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Canned tuna Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Fresh tuna Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Dry salted cod (wet) Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Horse Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Fish douradines Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Espadarte Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Fish eggs Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Roasted sardines (in the coal) Monitor	Smoked/carbonized meat (HAP, heterocyclic amines)	GSTM1, GSTA1, NAT2	Weakest II phase detoxification for HAP and heterocyclic amines	Avoid carbonization; marinate and cook at moderate temperature.	Weakness <i>chip + inference (GSTM1 is CNV)</i>
Canned sardine Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Surimi / Delightful Sea Monitor	Salt (genetic sensitivity to salt) Sucrose (sucrose-isomaltase)	AGT, ADD1, CYP11B2, GNB3, ACE; SI	Moderate-high salt: counting daily · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Count in daily salt limit (< 5 g). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate-weak <i>chip</i>
Sushi / sashimi Monitor	Salt (genetic sensitivity to salt) Refined hydrates / high glycemic index	AGT, ADD1, CYP11B2, GNB3, ACE; TCF7L2, FTO, MTNR1B, FABP2	Moderate-high salt: counting daily · Intermediate glycemic response; worst at night (MTNR1B)	Count in daily salt limit (< 5 g). Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip</i>

3.11 Marisco and mollusks — 20 verified foods

limit: 1 · surveillance: 1 · without genetic predisposition: 18

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Dry shrimp Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Canned shellfish Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>

3.12 Eggs — 7 verified foods

no food with predisposition · without genetic predisposition: 7

3.13 Fats and oils — 19 verified foods

no food with predisposition · without genetic predisposition: 19

3.14 Non-alcoholic beverages (coffee, tea, cocoa, juices, soft drinks) — 35 verified foods

limit: 13 · surveillance: 14 · without genetic predisposition: 8

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Guaraná beverage Limit	Free sugar/fructose x liver Coffee Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; CYP1A2, ADORA2A; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Moderate caffeine: attention to total dose and time · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Count total caffeine of the day; avoid in the afternoon and at night. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Energy drink Limit	Coffee Free sugar/fructose x liver Sucrose (sucrose-isomaltase)	CYP1A2, ADORA2A; PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p89): dose and time dependent effect · High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Up to ~200 mg of caffeine per day and only in the morning. Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip</i>
Sugar-free energy drink Limit	Coffee	CYP1A2, ADORA2A	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p89): dose and time dependent effect	Up to ~200 mg of caffeine per day and only in the morning.	Moderate <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Smoked or boiled coffee (not filtered) Limit	Coffee	CYP1A2, ADORA2A	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p89): dose and time dependent effect	Up to ~200 mg of caffeine per day and only in the morning.	Moderate <i>chip</i>
Expressed coffee Limit	Coffee	CYP1A2, ADORA2A	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p89): dose and time dependent effect	Up to ~200 mg of caffeine per day and only in the morning.	Moderate <i>chip</i>
Filtered coffee Limit	Coffee	CYP1A2, ADORA2A	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p89): dose and time dependent effect	Up to ~200 mg of caffeine per day and only in the morning.	Moderate <i>chip</i>
Sugar iced tea Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Tea matcha Limit	Coffee	CYP1A2, ADORA2A	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p89): dose and time dependent effect	Up to ~200 mg of caffeine per day and only in the morning.	Moderate <i>chip</i>
Gas / limonada Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Powdered chocolate milk (chocolate) Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Glue refrigerator Limit	Free sugar/fructose × liver Coffee Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; CYP1A2, ADORA2A; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Moderate caffeine: attention to total dose and time · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Count total caffeine of the day; avoid in the afternoon and at night. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Orange / lemon Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Tonic water (quinine) Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Isotonic beverage Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Cocoa powder Monitor	Coffee Food Histamine	CYP1A2, ADORA2A; AOC1 (DAO), HNMT	Moderate caffeine: attention to total dose and time · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Count total caffeine of the day; avoid in the afternoon and at night. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate <i>chip</i>
Instant coffee Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate <i>chip</i>
Cappuccino Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate <i>chip</i>
Beer without alcohol Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Hot chocolate Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Food Histamine	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; AOC1 (DAO), HNMT	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate <i>chip + imputation</i>
Mate tea Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate <i>chip</i>
Tea oolong Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate <i>chip</i>
Black tea Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate <i>chip</i>
Green tea Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate <i>chip</i>
Galon / coffee with milk Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate <i>chip</i>
Infusion of regaliz (alcaçuz) Monitor	Liquorice/regaliz (glycirrhzal)	CYP11B2, AGT, ADD1	Glycirrhzin retains sodium; more relevant with salt sensitivity	Avoid the regular consumption of licorice/regaliz.	Weak <i>chip</i>
Kombucha Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Glue refrigerator without sugar Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate <i>chip</i>

3.15 Alcoholic beverages — 16 verified foods

limit: 16 · without genetic predisposition: 0

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Brandy / bagasse Limit	Alcohol	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7	Normal acute tolerance, but higher hepatic susceptibility to alcohol	Zero alcohol.	Moderate-strong <i>chip + imputation</i>
Beer Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong <i>chip + imputation</i>
Sweetened cocktails Limit	Alcohol Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Normal acute tolerance, but higher hepatic susceptibility to alcohol · High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Zero alcohol. Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate-strong <i>chip + imputation</i>
Foaming / cleanliness Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong <i>chip + imputation</i>
Gin Limit	Alcohol	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7	Normal acute tolerance, but higher hepatic susceptibility to alcohol	Zero alcohol.	Moderate-strong <i>chip + imputation</i>
Licores (gink, bitter almond) Limit	Alcohol Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Normal acute tolerance, but higher hepatic susceptibility to alcohol · High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Zero alcohol. Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate-strong <i>chip + imputation</i>
Rum Limit	Alcohol	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7	Normal acute tolerance, but higher hepatic susceptibility to alcohol	Zero alcohol.	Moderate-strong <i>chip + imputation</i>
Sacqué Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong <i>chip + imputation</i>
Sidra Limit	Alcohol Food Histamine Free sugar/fructose × liver	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT; PNPLA3, MBOAT7, SERPINA1, HSD17B13	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate free sugar: dose-dependent hepatic effect	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Moderate portion; counting in the total daily free sugar.	Moderate-strong <i>chip + imputation</i>
White wine Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong <i>chip + imputation</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Rosé wine Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong <i>chip + imputation</i>
Red wine Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong <i>chip + imputation</i>
Green wine Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong <i>chip + imputation</i>
Generous / fortified wines Limit	Alcohol Food Histamine Free sugar/fructose × liver	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT; PNPLA3, MBOAT7, SERPINA1, HSD17B13	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate free sugar: dose-dependent hepatic effect	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Moderate portion; counting in the total daily free sugar.	Moderate-strong <i>chip + imputation</i>
Vodka Limit	Alcohol	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7	Normal acute tolerance, but higher hepatic susceptibility to alcohol	Zero alcohol.	Moderate-strong <i>chip + imputation</i>
Scotch Limit	Alcohol	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7	Normal acute tolerance, but higher hepatic susceptibility to alcohol	Zero alcohol.	Moderate-strong <i>chip + imputation</i>

3.16 Fermented and preserved fermented — 13 verified foods

limit: 7 · surveillance: 5 · without genetic predisposition: 1

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Canned chucrutz Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Beer yeast / yeast extract Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Fermented pepper mass Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Fish sauce (nam pla) Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Soybean sauce Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Fermented pickles (lacto-fermented) Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Garlic and pepper fermented seasoning (for meat) Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Water Kefir Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Mother-mass bread Monitor	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Moderate-high salt: counting daily · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Count in daily salt limit (< 5 g). Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Balsamic wine Monitor	Food Histamine Free sugar/fructose × liver	AOC1 (DAO), HNMT; PNPLA3, MBOAT7, SERPINA1, HSD17B13	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate free sugar: dose-dependent hepatic effect	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Moderate portion; counting in the total daily free sugar.	Weak <i>chip</i>
Cider vinagre Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>
Wine wine Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>

3.17 Sugars, sweeteners and sweets — 30 verified foods

limit: 20 · surveillance: 2 · without genetic predisposition: 8

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Sugarcane (regaliz sweet) Limit	Free sugar/fructose × liver Licorice/regaliz (glycirrhzal) Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; CYP11B2, AGT, ADD1; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Glycirrhzin retains sodium; more relevant with salt sensitivity · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Avoid the regular consumption of licorice/regaliz. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Sweet cotton Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Yellow/masked sugar Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
White sugar Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Whole sugarcane sugar (grape) Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Bombos Limit	Free sugar/fructose × liver Food Histamine Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; AOC1 (DAO), HNMT; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
White chocolate Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Milk chocolate Limit	Free sugar/fructose × liver Food Histamine Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; AOC1 (DAO), HNMT; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Milk sweet Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Eggs/mole eggs Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Powder fructose Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate <i>chip + imputation</i>
Sugared gelatin Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Marmalade / quince jelly Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Mel Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate <i>chip + imputation</i>
Sugarcane molasses Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Nougat / roast Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Sweeteners/gums Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Agave syrup Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate <i>chip + imputation</i>
Glucose-fructose syrup (HFCS) Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate <i>chip + imputation</i>
Maple syrup Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Black chocolate 70% Monitor	Coffee Food Histamine	CYP1A2, ADORA2A; AOC1 (DAO), HNMT	Moderate caffeine: attention to total dose and time · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Count total caffeine of the day; avoid in the afternoon and at night. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate <i>chip</i>
Chocolate mousse Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Food Histamine	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; AOC1 (DAO), HNMT	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate <i>chip + imputation</i>

3.18 Spices, herbs, condiments and sauces — 45 verified foods

limit: 7 · surveillance: 12 · without genetic predisposition: 26

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Concentrated broth (cube) Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Meat extract Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Malagueta mass Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Worcestershire Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Barbecue sauce Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose · Moderate-high salt: counting daily · Weakest II phase detoxification for HAP and heterocyclic amines	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose. Count in daily salt limit (< 5 g). Avoid carbonization; marinate and cook at moderate temperature.	Moderate <i>chip + imputation</i>
Cooking salt Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak <i>chip</i>
Sea salt / salt flower Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak <i>chip</i>
Dry coriander seeds Monitor	Corianders (soap flavor)	OR6A2	A slightly higher probability of finding the corianders with flavor (only flavor)	It only affects the taste; without implication for health.	Weak <i>chip</i>
Monosodium glutamate Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Pyri-pyri sauce in oil Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Soybean sauce reduced in salt Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Industrial tomato sauce Monitor	Food Histamine Free sugar/fructose × liver Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; PNPLA3, MBOAT7, SERPINA1, HSD17B13; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate free sugar: dose-dependent hepatic effect · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Moderate portion; counting in the total daily free sugar. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Hot sauce (tabasco, sriracha) Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Mustard (wet) Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Smoked paprika Monitor	Smoked/carbonized meat (HAP, heterocyclic amines)	GSTM1, GSTA1, NAT2	Weakest II phase detoxification for HAP and heterocyclic amines	Avoid carbonization; marinate and cook at moderate temperature.	Weakness <i>chip + inference (GSTM1 is CNV)</i>
Pestus Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Liquorice root (cattlewood) Monitor	Liquorice/regaliz (glycirrhizal)	CYP11B2, AGT, ADD1	Glycirrhizin retains sodium; more relevant with salt sensitivity	Avoid the regular consumption of licorice/regaliz.	Weak <i>chip</i>
Green parsley / chimichurri Monitor	Corianders (soap flavor)	OR6A2	A slightly higher probability of finding the corianders with flavor (only flavor)	It only affects the taste; without implication for health.	Weak <i>chip</i>
Vinagret Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak <i>chip</i>

3.19 Pastry, desserts and sweet snacks — 24 verified foods

limit: 19 · surveillance: 4 · without genetic predisposition: 1

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Camel Baba Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Butterball Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Chocolate cake Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Fried cake filled with cream Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Bolo-Rei Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Cheesecake Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Donut Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Burned cream Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Muffin Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Palmier Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Pastel de nata Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Pudim flan Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Ló bread Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Queijada Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Horseradish Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Dreams / children Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Fruit ice cream Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Tiramisu Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Pillow / sweet leaf Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Cereal bar Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Oats Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Apple pie Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>
Waffles / sweet crepes Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate <i>chip + imputation</i>

3.20 Ready meals, fast food and processed salty foods — 39 verified

limit: 14 · surveillance: 23 · without genetic predisposition: 2

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Sugar-tuffed rice Limit	Free sugar/fructose × liver Refined hydrates / high glycemic index Sucrose (sucrose-isomaltase) Fortified foods (iron, folic acid)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; TCF7L2, FTO, MTNR1B, FABP2; SI; HFE, MTHFR	High doses of free sugar/fructose increase liver fat in patients with 148M · Intermediate glycemic response; worst at night (MTNR1B) · Activity of saccharase slightly lower; possible digestive discomfort with much sucrose · Iron added and synthetic folic acid: prefer natural sources	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Prefer integral versions, moderate portions, with protein or fiber and early in the day. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer versions without iron or folic acid added.	Moderate <i>chip + imputation</i>
Meat and sausage cooking Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Bean Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
France Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Fast food burger Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak <i>chip</i>
Kebab/shawarma Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak <i>chip</i>
Instant mass (noodles) Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak <i>chip</i>
Pepperoni pizza Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Pretzels Limit	Salt (genetic sensitivity to salt) Refined hydrates / high glycemic index	AGT, ADD1, CYP11B2, GNB3, ACE; TCF7L2, FTO, MTNR1B, FABP2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Intermediate glycemic response; worst at night (MTNR1B)	Small and infrequent portions; prefer versions with less salt. Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip</i>
Bread with sausage Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Ham sandwich Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Salted corn snacks Limit	Salt (genetic sensitivity to salt) Refined hydrates / high glycemic index	AGT, ADD1, CYP11B2, GNB3, ACE; TCF7L2, FTO, MTNR1B, FABP2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Intermediate glycemic response; worst at night (MTNR1B)	Small and infrequent portions; prefer versions with less salt. Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak <i>chip</i>
Instantaneous soup in package Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Strips of bacon / towers Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>
Rice of shellfish Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Duck rice Monitor	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Moderate-high salt: counting daily · Weakest II phase detoxification for HAP and heterocyclic amines	Count in daily salt limit (< 5 g). Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Cod with creams Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Bras Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Fast food potatoes Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak <i>chip + imputation</i>
Biphane (pig sandwich) Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Burrito / batos Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Fish caulder Monitor	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Moderate-high salt: counting daily · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Count in daily salt limit (< 5 g). Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Curry with rice Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Salted pastries/pastries Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Beans in can with sauce Monitor	Salt (genetic sensitivity to salt) Free sugar/fructose × liver	AGT, ADD1, CYP11B2, GNB3, ACE; PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate-high salt: counting daily · Moderate free sugar: dose-dependent hepatic effect	Count in daily salt limit (< 5 g). Moderate portion; counting in the total daily free sugar.	Moderate-weak <i>chip</i>
Chicken pyri Monitor	Smoked/carbonized meat (HAP, heterocyclic amines) Salt (genetic sensitivity to salt)	GSTM1, GSTA1, NAT2; AGT, ADD1, CYP11B2, GNB3, ACE	Weakest II phase detoxification for HAP and heterocyclic amines · Moderate-high salt: counting daily	Avoid carbonization; marinate and cook at moderate temperature. Count in daily salt limit (< 5 g).	Weakness <i>chip + inference (GSTM1 is CNV)</i>
Infant formula / dairy papayas Monitor	Fortified foods (iron, folic acid)	HFE, MTHFR	Iron added and synthetic folic acid: prefer natural sources	Prefer versions without iron or folic acid added.	Weak <i>chip</i>
Lasanha Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Chicken nuggets Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Paella Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Pizza margherita Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Prego (sands of steak) Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Frozen ready meal Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Rissols / croquettes / cod pastes Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>
Canned tuna salad Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak <i>chip</i>
Cheese sandwich Monitor	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Moderate-high salt: counting daily · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Count in daily salt limit (< 5 g). Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak <i>chip</i>
Mixed toast Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak <i>chip</i>

Abstract: 324 of the 626 verified foods have no genetic predisposition and are not listed.

4. Limitations

- **Genetics only explains part of the intolerances.** Most reactions to food depend on the intestine, microbiome, dose, preparation and other non-genetic factors.
- **IgE allergies** (brown, dried fruits, shellfish, fish, egg, milk, sesame, etc.) are not predictable by genetics: IL4R Q576R heterozygous, the possible loss of filagrin function (unconfirmed) and the loci HLA give only atopic context. The dietary IgG tests do not serve to decide evictions.
- **FODMAP, polyols and fructose malabsorption** (sorbitol, xylitol, fructose in excess, legumes) and the sensitivity to non-celiac gluten do not have a validated genetic test; they are non-determinable.
- **Chip + imputation, no sequencing.** Only the variants measured or imputed with $DR2 \geq 0.8$ are seen; rare ($< 0.1\%$) and new variants are not seen. A result "without predisposition" means "without the variants tested", not "without variants". Not measurable: AMY1 (amylase, CNV), TPSAB1 (tryptase, CNV), GSTM1/GSTT1 (inferred), repetitions and high resolution HLA.
- **Dose effects:** orange and yellow indicate dose- and frequency-dependent effects (sugar, salt, caffeine, alcohol, histamine), neither allergy nor mandatory intolerance. Colors assume usual portions.
- **Evidence:** histamine (DAO/HNMT), saccharase (SI Val15Phe), coriander, iron (H63D) and smoked (GSTM1/GSTA1/NAT2) are based on association studies with small effects. Lactase, HLA-DQ celiac, ALDOB, G6PD and PAH have strong evidence.
- **List of foods:** results from a selection based on statistics and composition tables; the real composition (salt, sugar, histamine) varies by brand, revenue and maturation.
- **It is not a diagnosis.** If there are symptoms, confirm with clinical tests: hydrogen respiratory test (lactose, fructose, sucrose), tTG-IgA with total IgA (celiac disease, still eating gluten), allergy tests IgE with the allergologist and, for histamine, medical evaluation with a food diary. All indications are suggestion (not applied).

5. Sources and methods

- List of foods: products from FAO Food Balance Sheets (FAOSTAT, 2010–2023), tables of food and food composition, dishes and processed frequent consumption worldwide; manual deduplication and classification in 21 categories.
- Genotypes: genotyping chip (GRCh38) as primary source; Beagle 5.5 imputation with the full genome reference panel 30x (only $DR2 \geq 0.8$); HLA tags calibrated in the reference population.
- ClinVar (28-09-2026 version): Scanning of pathogenic/probably pathogenic variants in ~130 food metabolism genes (ALDOB, SI, LCT, PAH, AGXT, GALT, G6PD, AOC1, HNMT, HFE, urea cycle, among others).
- Background: persistence of lactase (Enattah 2002); HLA-DQ and celiac disease (Monsuur 2008; ESPGHAN); ALDOB (Cross 1990; Santer 2005); SI and IBS (Henström 2018); DAO/HNMT (Maintz 2011; García-Martin 2015); CYP1A2/ADORA2A (Cornelis 2006; Rétey 2007; Cornelis 2011); salt (consensus AHA on sensitivity to salt 2016); PNPLA3 × sugar and alcohol (Davis 2010; Sevastianova 2011; Buch 2015); APOA2 (Corella 2009); GSTM1/NAT2 and cooked meat (IARC 2018); OR6A2 (Eriksson 2012); G6PD (OMS); PAH (ClinVar).